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upper - lower
unit

Q2/ex2.

$$\int \frac{dn}{n^2 \sqrt{n^2 + 1}}$$

sol

$$n = a \tan \theta$$

$$\sqrt{n^2 + a^2} = n \sec \theta$$

$$n = (1)^2 \tan \theta \checkmark$$

$$n = \tan \theta$$

$$dn = \sec^2 \theta d\theta$$

Now

$$\int \frac{\sec^2 \theta d\theta}{n^2 \sqrt{(1)^2 + \tan^2 \theta}}$$

$$\int \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sqrt{1 + \tan^2 \theta}}$$

$$\int \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sqrt{1 + \tan^2 \theta}}$$

$$\text{Note: } \tan^2 \theta + 1 = \sec^2 \theta$$

$$\int \frac{\sec^2 \theta d\theta}{\tan^2 \theta \sqrt{\sec^2 \theta}}$$

$$= \tan^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta} \text{ so } \sin^2 \theta \times \frac{1}{\cos^2 \theta}$$

$$= \sin^2 \theta \cdot \sec^2 \theta$$

because $\frac{1}{\cos^2 \theta} = \sec^2 \theta$.

$$= \int \frac{\sec^2 \theta \, d\theta}{\tan^2 \theta \sqrt{\sec^2 \theta}}$$

$$= \int \frac{\cancel{\sec^2 \theta} \, d\theta}{\sin^2 \theta \cdot \cancel{\sec^2 \theta} \cdot \sec \theta}$$

$$= \int \frac{d\theta \cos \theta}{\sin \theta \sin \theta}$$

Note $\frac{\cos \theta}{\sin \theta} = \cot \theta$

$$= \int \cot \theta \times \frac{1}{\sin \theta}$$

$$\int \cot \theta \operatorname{cosec} \theta \, d\theta$$

integrals

$$= -\operatorname{cosec} \theta + C$$

in terms of x

$$= \frac{-\sqrt{x^2 + 1}}{x} + C$$

Ans